INTRODUCRION TO HYPER-V

What are hypervisors?

- A hypervisor, also known as a virtual machine monitor or VMM, is software that creates and runs virtual machines (VMs). A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.

- Type of hypervisors:

There are two types of hypervisors: Type 1 and Type 2 hypervisors. Both hypervisor varieties can virtualize common elements such as CPU, memory and networking, but based on its location in the stack, the hypervisor virtualizes these elements differently.

A hypervisor enables the host hardware to operate multiple VMs independent of each other and share abstracted resources among those VMs. Virtualization with a hypervisor increases a data center's efficiency compared to physical workload hosting.

- **Type 1 hypervisors**

A Type 1 hypervisor runs directly on the host machine's physical hardware, and it's referred to as a bare-metal hypervisor. The Type 1 hypervisor doesn't have to load an underlying OS. With direct access to the underlying hardware and no other software -- such as OSes and device drivers -- to contend with for virtualization, Type 1 hypervisors are regarded as the most efficient and best-performing hypervisors available for enterprise computing.

Hypervisors that run directly on physical hardware are also highly secure. Virtualization mitigates the risk of attacks that target security flaws and vulnerabilities in OSes because each guest has its own OS. This ensures an attack on a guest VM is logically isolated to that VM and can't spread to others running on the same hardware.

IT organizations use Type 1 hypervisors for production-level workloads that require increased uptimes, advanced failover and other production-ready features. The typical Type 1 hypervisor can scale to virtualize workloads across several terabytes of RAM and hundreds of CPU cores.

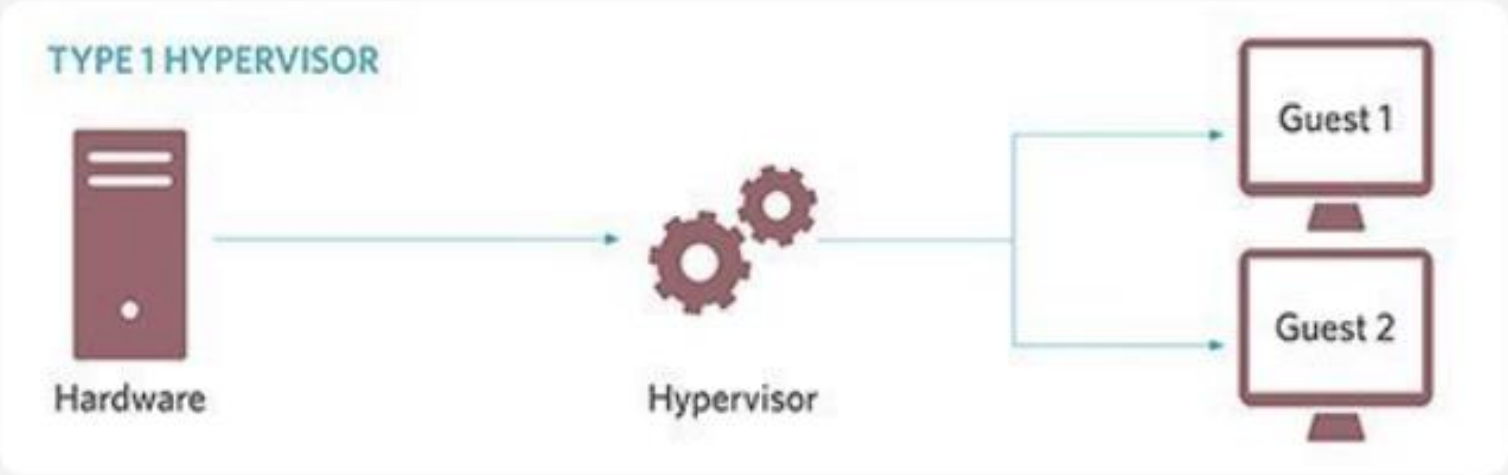
In addition, Type 1 hypervisors often provide support for software-defined storage and networking, which creates additional security and portability for virtualized workloads. However, such features come with a much higher initial cost and greater support contract requirements.

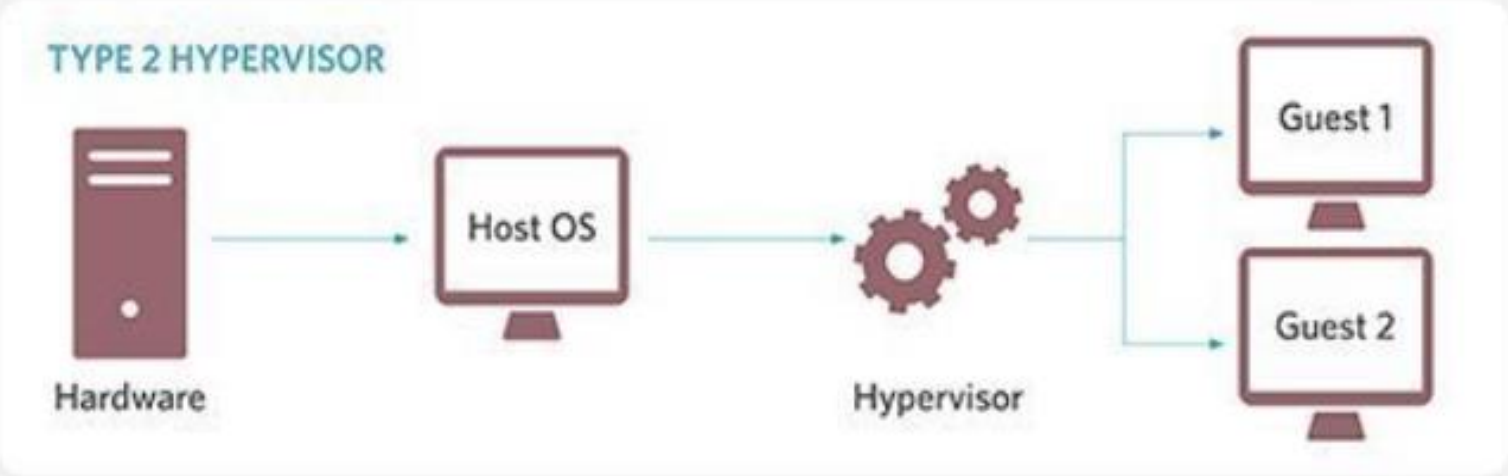
- **Type 2 hypervisors**

A Type 2 hypervisor is typically installed on top of an existing OS. It is sometimes called a hosted hypervisor because it relies on the host machine's preexisting OS to manage calls to CPU, memory, storage and network resources.

Consequently, Type 2 hypervisors are generally not used for data center computing and are reserved for client or end-user systems -- sometimes called client hypervisors -- where performance and security are lesser concerns. They also come at a lower cost than Type 1 hypervisors and make an ideal test platform compared to production virtualized environments or the cloud. For example, software developers might use a Type 2 hypervisor to create VMs to test a software product prior to release. IT organizations typically use Type 2 hypervisors to create virtual desktops. Type 2 hypervisors can support large and complex nested environments

Hypervisor types





Virtualization has solved our resource utilization problems, by enabling us to make more effective use of the hardware available. This means that we can now host more workloads using less space inside of our datacenters. We can dynamically move workloads from one piece of physical hardware to another, without any interruption to the running workload. We can even provide high-availability functionality to workloads using these virtualization technologies. It's not difficult to see how virtualization is changing the IT management landscape. System administrators should now be looking to add this knowledge to their toolbox, to help them manage ever-increasing IT complexity, as virtualization continues to be a staple of IT departments everywhere. While there are many different virtualization technologies on the market today, this course will focus on Microsoft's Hyper-V, with a primary focus on Windows Server 2019 and Windows Server 2016.

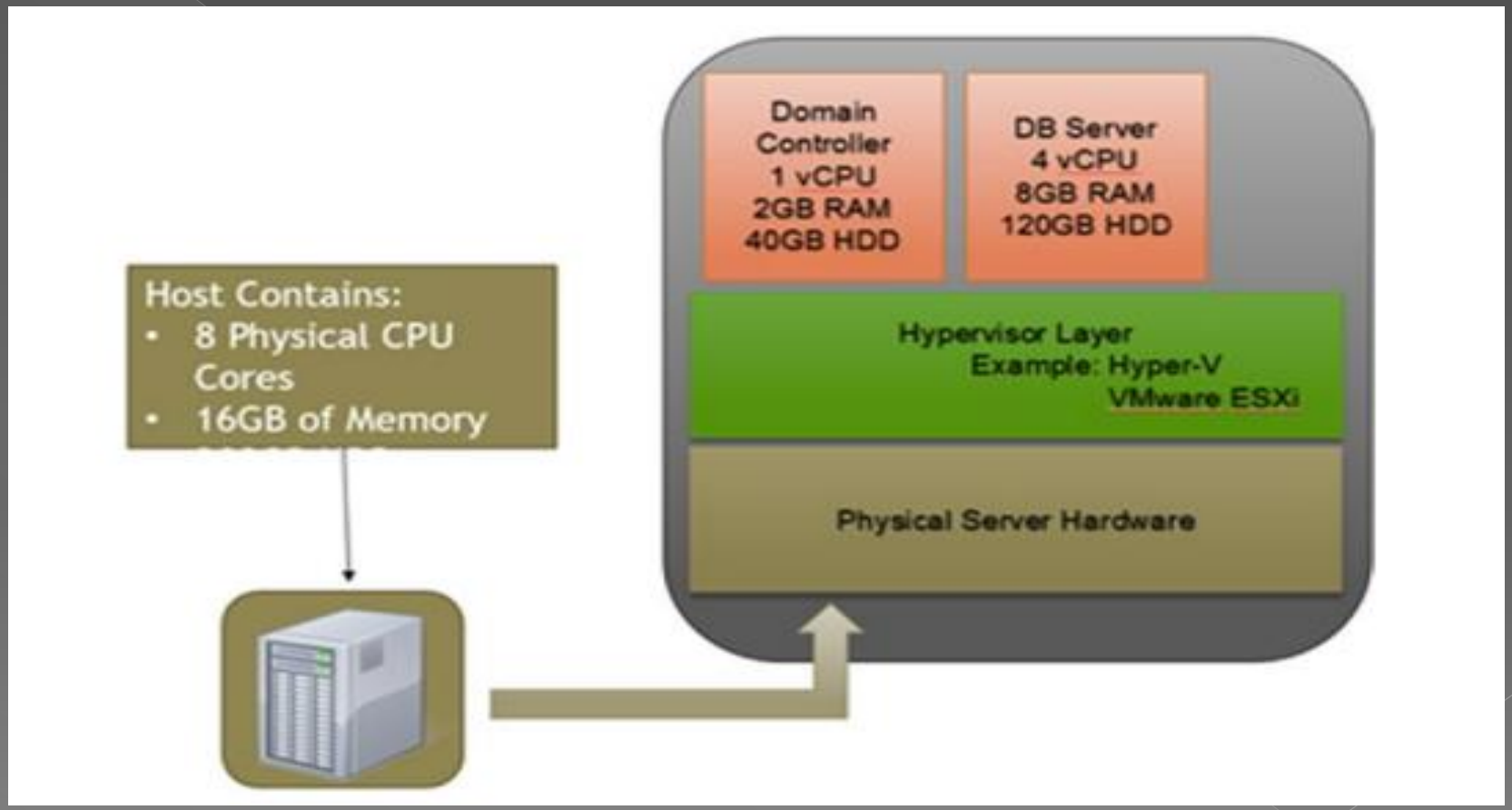
I- INTRODUCTION TO HYPER-V

- where this technology could fit in your organization. Once we understand the "why," we'll explain how this technology works from the ground up. We'll continue to build on this foundation throughout this course. Let's dig in.

What Is Hyper-V?

Simply put, Hyper-V is a hypervisor. The problem with this answer is that it will likely lead you to the question "What is a hypervisor?" A hypervisor is a piece of software that is designed to delegate access to physical resources, such as CPU and memory, to multiple virtualized operating systems (OSs) or applications, as shown in Figure 1-1.

Figure 1-1. Logical layout of a hypervisor


Domain Controller
1 vCPU
2GB RAM
40GB HDD
DB Server
4 vCPU
8GB RAM
120GB HDD
Host Contains:
• 8 Physical CPU Cores
• 16GB of Memory
Hypervisor Layer
Example: Hyper-V
VMware ESXi
Physical Server Hardware

In Figure 1-1, we have a physical server that contains 8 physical CPUs and 16GB of memory. The hypervisor layer gets installed directly on top of the physical hardware and acts as a broker for the resources contained in the host machine.

Figure 1-1 shows that two “virtualized” servers have been created, and each is being given a portion of the host’s physical resources. Each of these virtual machines (VMs) contains fully functional Figure 1-1. Logical layout of a hypervisor Chapter 1 Introduction to Hyper-V NEWOUTLOOK.IT https://t.me/nettrain 3 OSs and will act just like physical servers on your network.

They are also isolated and separated from one another, just like physical servers. This is the most basic function of a hypervisor such as Hyper-V

II- The Business Benefits of Hyper-V

- Hypervisors have been around for some time. Widespread adoption started more than ten years ago, when IT professionals saw the inherent benefits of virtualized workloads. Some variant of VMware's ESXi product has been around for the last ten years or so, and some mainframe systems have been utilizing this type of technology since the 1980s!

- Let's look at this from a business perspective, using the example of the fictional company XYZ Corp., which is approaching a hardware refresh cycle. This approach will enable you to see why the industry has transitioned to virtualized workloads.

Setting the Stage

XYZ Corp. is a company that is getting ready to make some expenditures on technology as part of a three-year refresh cycle. Its existing infrastructure has been in place for some time, and no virtualization technologies are in use today. Cost is a major concern, as well as stability. The IT department has been given a limited budget for upgrades, but several of the physical machines are out of support and warranty, and a significant investment would be needed to replace each existing physical machine with another, more current model.

III- The Physical Computing Model

- To date, XYZ Corp. has used the older computing model of one role/function per physical server. This methodology has a number of flaws associated with it. Before we go any further, let's get a look at what it currently has. The servers and workloads owned and operated by XYZ are shown in Figure 1-2.

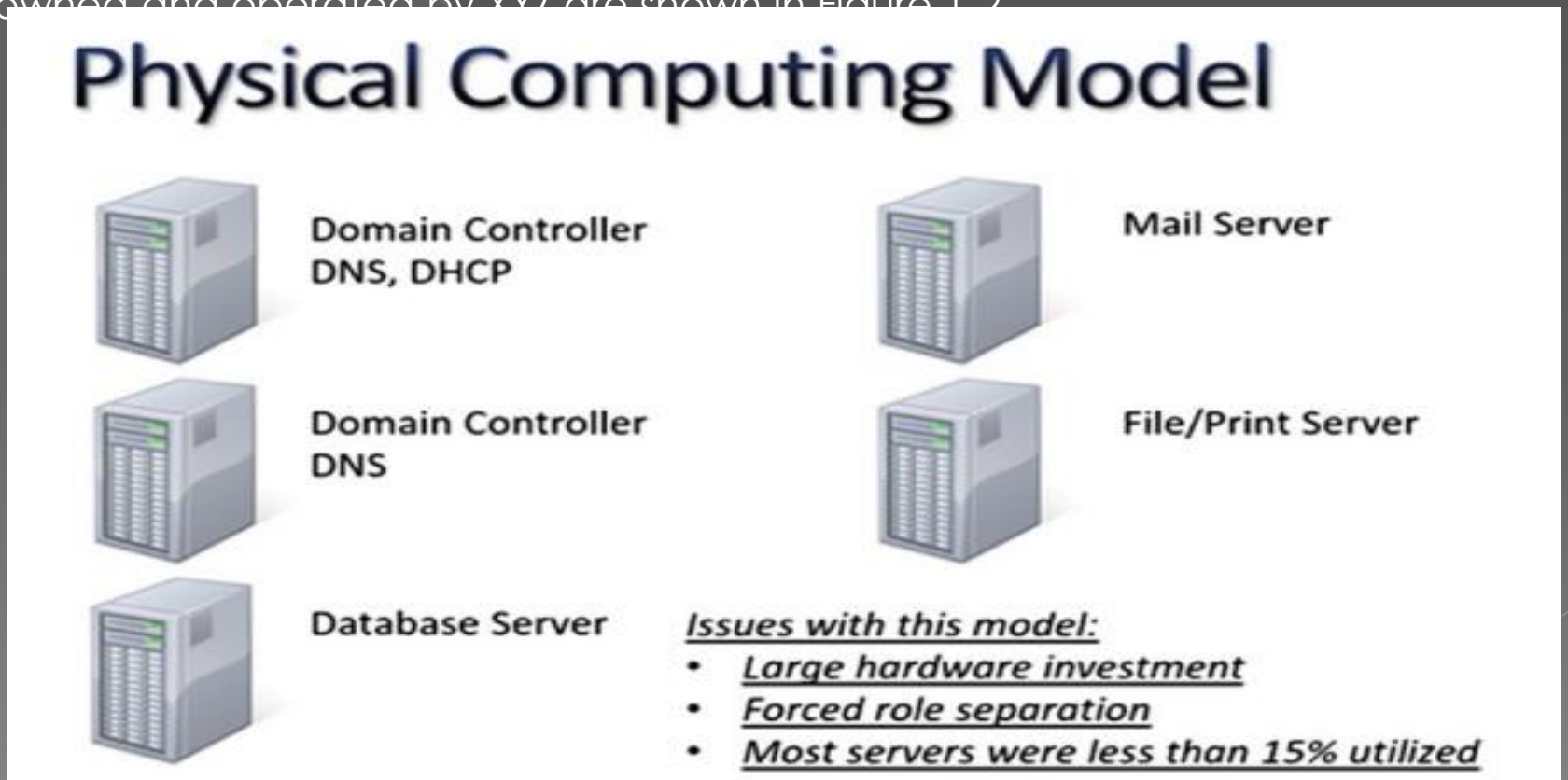


Figure 1-2. *A list of physical servers owned and operated by XYZ Corp*

- Figure 1-2 shows that XYZ Corp. has five physical servers and that each server generally follows the rule of one role/feature per physical box. Sadly, this method was not conducive to efficient use of computing resources. Historically, most systems would go through life running at only 20% utilization (on average), leaving the remaining 80% of the system with nothing to do. One role per server was never an official standard; it was an unspoken rule that most system admins adopted. In reality, there are so many bits of software on any given system that, once in a while, issues are bound to arise wherein some of that code doesn't play nicely with other code on the system. This can cause system outages or degrade performance. Downtime equals money lost to the business. The unspoken rule of one role per machine helped prevent this from occurring.

Before the widespread introduction of virtualization, many organizations invoked projects to reduce the number of servers in their environment. This was essentially a cost-cutting exercise, in order to free up hardware for new applications. Consolidation projects would attempt to match workloads, to enable multiple applications to run on the same server, with varying levels of success. If your management suggests that a server consolidation project would be a good thing to attempt, you should immediately start talking up the benefits of virtualization. Virtualization gives you the benefits of server consolidation but retains the application isolation inherent in the one-application-per-server concept.

The increase in cost to buy additional physical servers to fill all the roles needed on the network was an acceptable cost, because at the time, there wasn't a mainstream technology that would allow system admins to make better use of the available hardware. Nor was there any software available that would ensure the separation of workloads to prevent those situations in which two pieces of code are in conflict and create issues. Virtualization technology has now solved this computing dilemma, in the form of the hypervisor.

IV-Making the Change to Virtualized Computing

- Virtualization technology enables XYZ Corp. to purchase one or two new, powerful physical servers and place the workloads shown in Figure 1-2 into isolated "virtual" instances on top of a hypervisor running on the new equipment. More powerful servers will be more expensive than a similar though less powerful system, but reducing the number of physical servers and maintaining warranty and support on them usually allows you to have a much lower overall cost.

Table 1-1 reviews the resources that are currently in use by the physical workloads.

Table 1-1. *Total Physical Computing Resources Configured for XYZ Corp*

Workload	Number of Physical CPU Cores	Amount of Physical Memory in GB	Amount of Storage in GB
Domain Controller	2	2	80
Domain Controller	2	2	80
Database Server	4	8	300
Mail Server	4	8	300
File/Print Server	2	4	500
Total	14 CPU Cores	24 GB	1260 GB

- **Note** : You can use such tools as Perform, Task Manager, and others to discover and capture this type of information.
- As an example, let's say that after examining the resource demand for each physical server, XYZ Corp. determines that the actual resource demand for each workload (with a little wiggle room) is as shown in Table 1-2.

Table 1-2. *Projected Resource Demand per Virtualized Workload for XYZ Corp*

Workload	Number of Physical CPU Cores	Amount of Physical Memory in GB	Amount of Storage in GB
Domain Controller	1	2	50
Domain Controller	1	1	50
Database Server	2	6	200
Mail Server	2	6	150
File/Print Server	1	2	250
Total	7 CPU Cores	17GB of RAM	700GB

Comparing Table 1-2 to Table 1-1 shows that a reduction of 7 CPU cores and 7GB of RAM are required, if the workloads are virtualized, and 560GB of storage space will be potentially released.

V-MORE TO LEARN

You may find that a server appears to be using a very small portion of the resources allocated to it, and you'll want to downsize the workload accordingly, once virtualized. You still need to take into account the minimum software requirements for the OS and the software running inside the virtualized OS, to maintain a supported configuration with the associated vendor.

For example, you may have a Microsoft SQL Server system that has a memory demand of 256MB of memory (unlikely, but it could happen). While you could size the workload in that manner once it's virtualized, the minimum system requirements for Microsoft SQL Server state that 1024MB of memory is the minimum required. Therefore, when sizing the memory for that workload, you wouldn't drop below 1024MB of memory for that VM.

- Without getting too into the details of the sizing math at this point, taking into account the resource requirements listed in Table 1-2 and adding the resource overhead required to run Hyper-V itself, XYZ Corp. now knows how powerful of a server (or servers) to procure. These new physical server(s) will be more powerful individually than the previous five physical servers and will conduct the work that all five of them performed previously.
- Additionally, it would be wise for XYZ Corp. to allocate an additional 20%–40% of resources to account for future growth. This is by no means an industry-defined standard. The needs of the organization will dictate this more than anything. Many organizations will add the maximum number of CPUs and maximize the RAM in a new virtualization host, as it's cheaper to do that than to add components at a later date. This approach ensures that there is room for future growth in the virtualized environment.

VI-The Virtualized Computing Model

Once the new hardware is purchased and arrives, Hyper-V will be installed and the workloads migrated in their existing state into a VM, or they may be migrated in a more traditional manner, if the OS and/or installed software is being upgraded as well. During this process, XYZ Corp. can allocate each of its core computing resources (CPU, memory, storage, and networking) to each virtualized workload, using the projected resource demand they determined earlier in Table 1-2.

- Once migration has been completed, each workload will be running on one of the new servers, each in its own isolated virtualized environment. In this new configuration, every entity on the XYZ Corp. network that was able to talk to the (now virtualized) workloads can still do so. Nothing has changed from a communications perspective. These workloads are still reachable on the network, and the end points won't even know that the workloads have been virtualized. Everything is business as usual for XYZ Corp.'s end users.
- Once XYZ Corp. has fully virtualized its computing workload, the server layout would look much like that shown in Figure 1-3. The physical server footprint has been reduced to two physical servers, with the five previously physical workloads now running as VMs. These workloads continue to function as though they are still physical boxes, which is the core function of our hypervisor—Hyper-V.

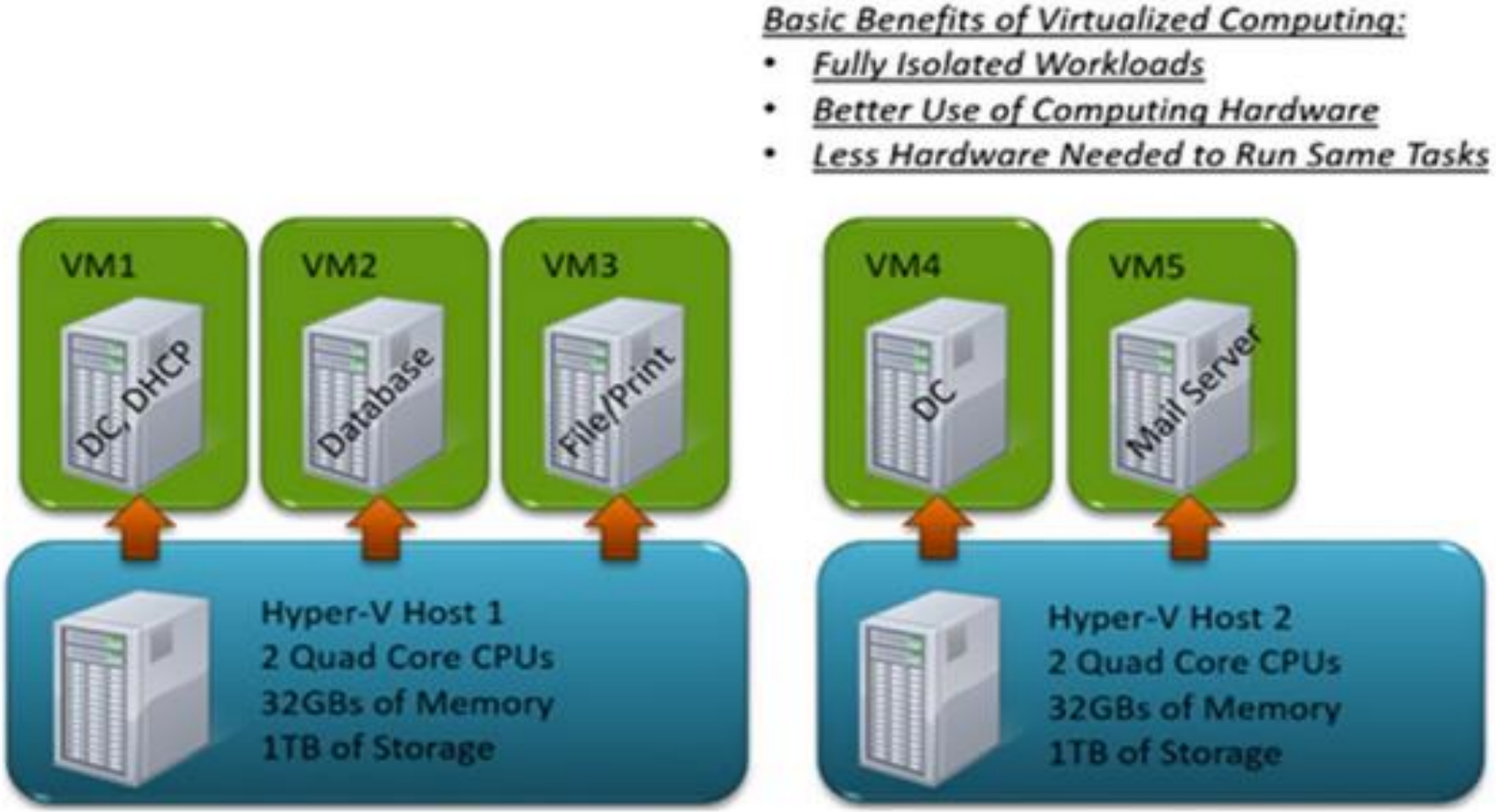


Figure 1-3. *XYZ Corp.'s now virtualized computing workload. The virtualization host's specifications are higher than the calculations from Table 1-2 would indicate, to allow for future growth of the environment.*

VII- Hyper-V Architecture

Figure 1-4 shows the Hyper-V architecture. Let's review what each component does in turn.

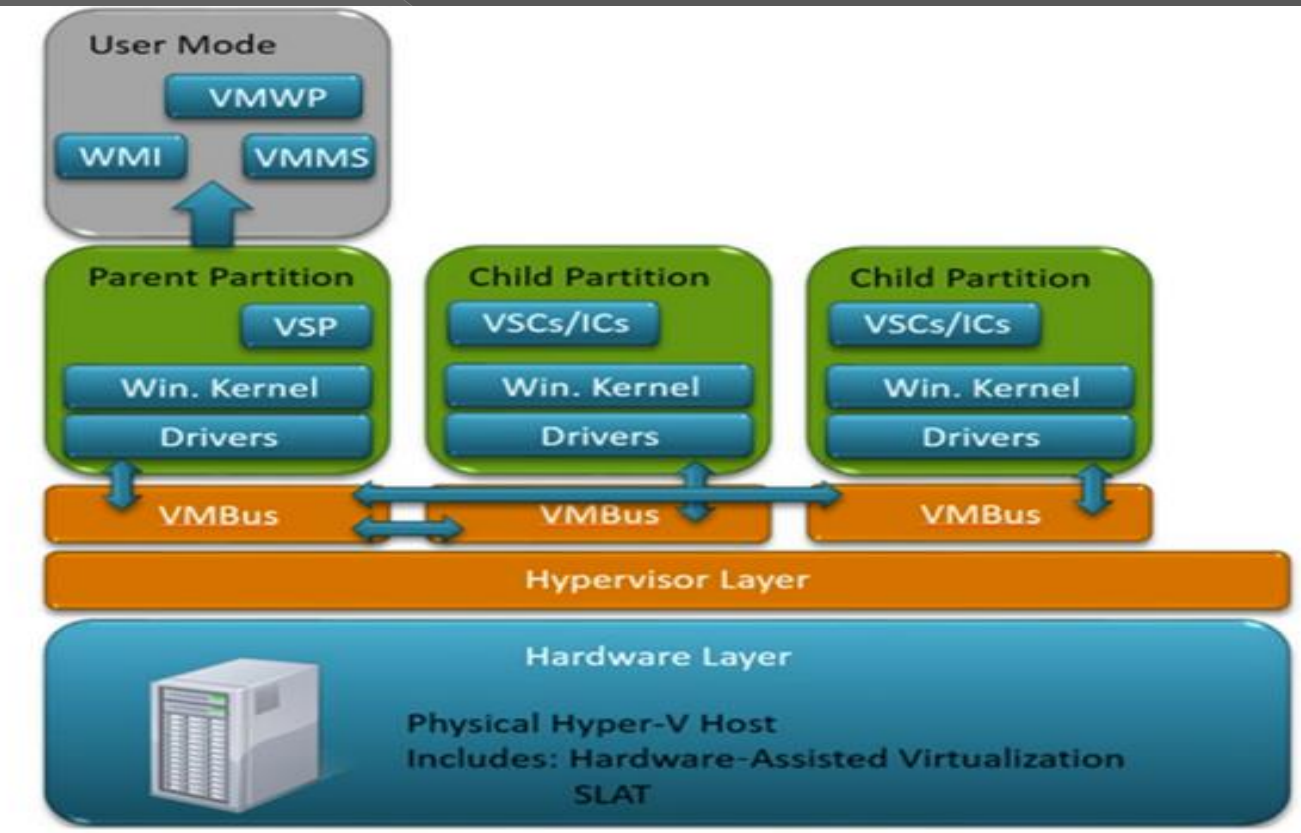


Figure 1-4. *High-level Hyper-V architecture*

VIII- Hardware Layer

- The hardware layer is the physical hardware on which Hyper-V is installed. There are some requirements that must be met:
- • Hyper-V expects that the machine will have hardware-assisted virtualization, such as Intel-VT or AMD-V.
- • SLAT (second-level address translation) is also a requirement. These are computing technologies that are contained within the CPU on your Hyper-V host. Hardware-assisted virtualization and SLAT both provide enhanced computational capabilities when dealing with virtualized workloads. Most new hardware supports these requirements, but some options may still have to be enabled in the systems BIOS. Check your system's documentation for details.

- **Hypervisor Layer**

The hypervisor controls access to CPU and memory and allocates them as resources to various partitions (or VMs) running on top of them. When the Hyper-V role is installed, the OS of the physical server is made virtual during the process and becomes the parent partition shown in Figure 1-4.

- **VMBus**

This is a virtual bus that enables communication between the host system and the VMs running on the host. If a virtual machine has the Hyper-V integration services installed, it will be able to utilize this bus for Information exchanges with the host.

- **Parent Partition**

This partition (or VM) is a "hidden" VM that acts as the management OS and controls the hypervisor. After you install the Hyper-V role on a Windows server, the hypervisor layer is placed between the hardware and the OS. The installed OS effectively becomes a VM running on Hyper-V, which is used to provide device drivers, power management services, storage, and networking functionality to the VMs running on the host.

- **Virtualization Service Provider (VSP)**

VSP is only in the parent partition. It provides support for virtualized devices with full driver capabilities contained within the child VMs over the VMBus, that is, it provides optimized access to hardware resources to child VMs.

- **Virtual Machine Worker Process (VMWP)**

The VM worker process (one separate process for each running VM on the Hyper-V host) provides management functionality and access from the parent partition to each of the child partitions. These processes run in user mode.

- **Virtual Machine Management Service (VMMS)**

The VMMS is responsible for managing the state of all virtual machines (or child partitions) running on the Hyper-V host. The VMMS runs in user mode, within the parent partition.

- **Child Partition**

Child partitions are the VMs hosted on your Hyper-V installation that are given access to CPU and memory by the hypervisor. Access to storage and networking resources is granted through the parent partition's VSP.

- **Virtualization Service Client (VSC)**

The VSC in each child VM utilizes the hardware resources that are provided by the parent partition. Communication is via the VMBus.

- **Integration Components (ICs)**

These components, also known as integration tools, allow the child partition to communicate with the hypervisor and provides a number of other functions and benefits.

the underlying architecture, and having an idea of how things are laid out, will help you comprehend why some functions and features in Hyper-V are set up and used the way they are.

Chapter 2- Configuring Hyper-V Host Settings

- We've been digging through introductions and theory for some time, so it's now time to put that to practical application. We'll start by configuring the host settings of a stand-alone Hyper-V host. Hyper-V Manager will be used to complete these tasks, as it provides all of the needed functionality to connect to hosts, create virtual machines (VMs), and manage them.
- **Note** A Hyper-V host may be referred to as a host or parent, while a VM may be referred to as a VM, guest, or child. The terms are interchangeable, and all will be seen throughout this course.

What Is Hyper-V Manager?

Hyper-V Manager is a Microsoft Management Console (MMC) snap-in that provides basic administrative capabilities for managing one or more stand-alone instances of Hyper-V.

- **Hyper-V Manager**

This is primarily designed to manage stand-alone hosts or multiple hosts that are not clustered. It's the entry-level Hyper-V management tool with which all beginners should start. We'll be using Hyper-V Manager for the next several chapters. This utility can be launched from Administrative Tools in the Windows Control Panel.

- **Failover Cluster Manager**

Failover Cluster Manager is similar in feel to Hyper-V Manager but offers more capabilities. It's designed to manage small clustered Hyper-V environments. If you have a clustered environment, this should be your default management tool for Hyper-V.

- **Windows Admin Center**

Windows Admin Center (WAC, previously known as Project Honolulu) is the new kid on the block when it comes to management tools for Hyper-V. WAC is a web-based tool that is designed to manage hyperconverged infrastructure (HCI) in the emerging cloud era. It's designed to manage hosts, VMs, storage, and more.

- **Connecting to Hyper-V Hosts**

Before you can manage your Hyper-V environment, you must first connect to the system with the Hyper-V role installed. On the client machine, you must have the Remote Server Administration Toolkit (RSAT) tools of the same or newer version installed, in order to connect to the intended Hyper-V Host.

Navigating the Hyper-V Manager UI

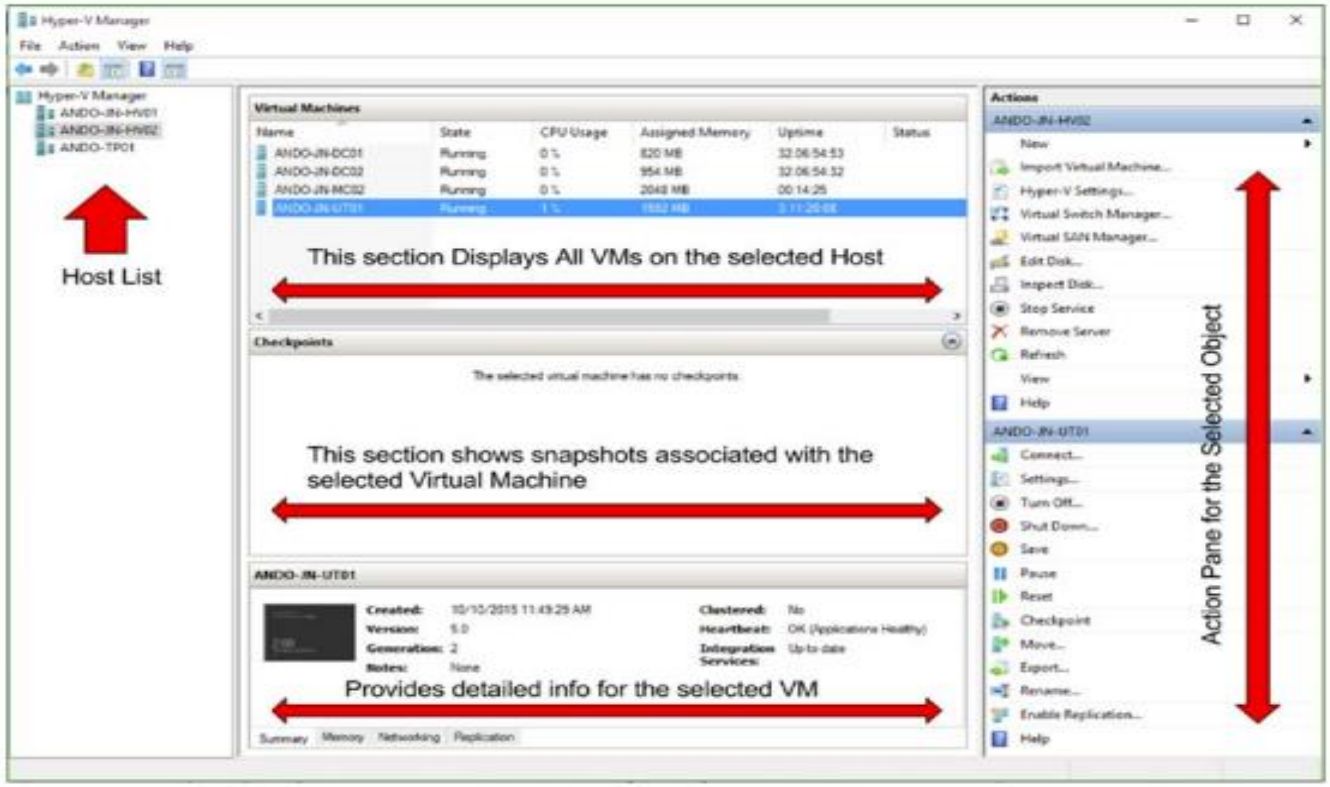


Figure 2-2. *Hyper-V Manager*

In Hyper-V Manager, as shown in Figure 2-2, hosts are listed on the left. VMs, their status, and any associated checkpoints (which we'll get into later) are listed in the center column, and actions associated with the selected object are listed in the right-hand column. The Hyper-V host in Figure 2-2 contains a number of running VMs. If you select one of them and review the details section near the bottom, you can see basic information and monitoring data for the VM. For example, in Figure 2-2, the VM ANDO-JN-UT01 is highlighted, and the summary tab has been selected. This tab gives us basic information about the VM, such as generation, version, and clustering status. The other tabs provide information about memory, networking, and replication settings, such as in Figure 2-3, which displays the memory summary for the VM ANDO-UB01. F

- When you first set up a new Hyper-V host, there are a number of items that should be configured before you start building VMs. It may also be necessary to modify some of these host settings in the future, if changes occur within your infrastructure. As an example, let's examine some of the basic settings.

1. Open Hyper-V Manager. Select the Hyper-V host in the left pane and click Hyper-V settings in the action pane. You'll see the dialog shown in Figure 2-4.

2. Make sure the Virtual Hard Disks section is selected on the left pane and select a location to store your virtual hard disks. You can pick a location or leave it as it is.

3. Select Virtual Machines in the left-hand pane and a location to store other VM related files. You can pick a location or leave it as it is.

4. Select the Live migration section and make sure that the maximum configured live migrations is 1. (For the purposes of a lab, no more is needed at this time.)

5. Click OK to close the window.

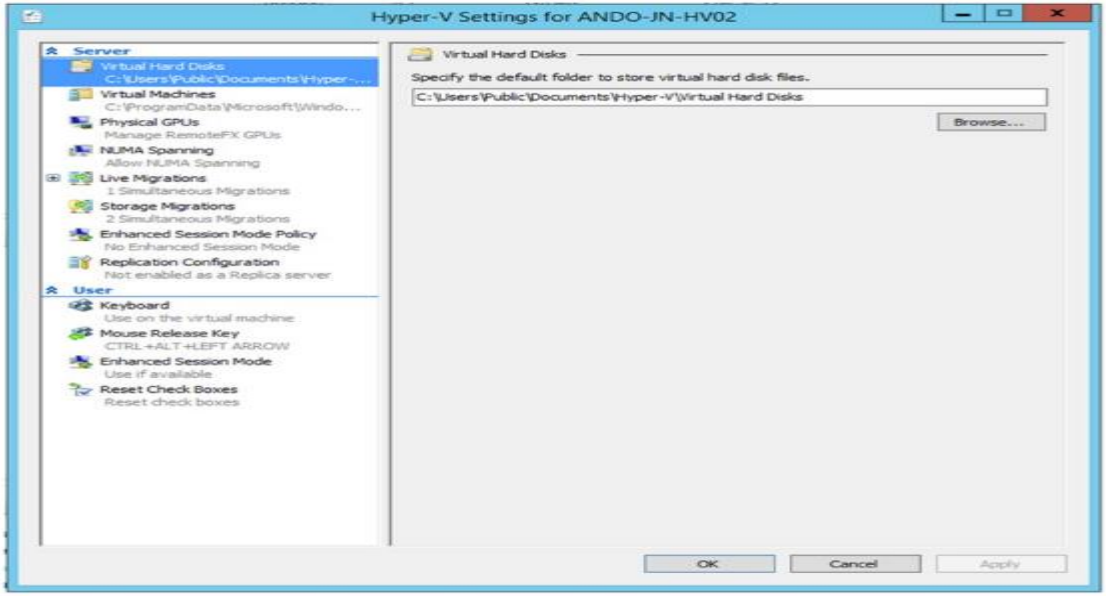


Figure 2-4. Hyper-V settings dialog

- Thus far, you've configured the storage location for your VMs and modified the live migration settings. There are many more options that can be configured in the UI. We'll briefly cover each item, and you can make further configuration changes in your lab, as needed. Please keep in mind that this is just a preview of most of the features mentioned. We will be covering each item in more detail as the course progresses.

- **Virtual Hard Disks**

Each VM has a virtualized disk drive in which to store files and the VM's operating system (OS). This information is stored in one of two formats. The older file format is VHD, primarily used by Windows Server 2008 R2 and older instances of Hyper-V. Newer Hyper-V versions use the VHDX variant by default, which provides enhanced features and added reliability. It is highly recommended that VHDX files be utilized whenever possible. In this section of the Hyper-V settings dialog, you can define the default storage location for virtual disk drives.

- **Virtual Machines**

This submenu controls the placement of VM components other than the virtual disks. The most important of these is the VM config file. It's the underlying file that Hyper-V references to keep track of a VM's configuration. It's quite common to see different locations for the VM and the virtual disks, as the VHDs and VHDXs can become quite large and may have to reside on their own volume or some kind of shared storage, as you'll see later.

- **Physical GPUs**

This section is mainly intended for use in virtual desktop infrastructure (VDI) scenarios. There are times when increased graphical performance provided by in-host graphics processing units (GPUs) and the Microsoft RemoteFX display protocol is required. This is where you configure this feature.

- **Live Migrations**

Live migration is a technology that enables you to move a VM from one physical Hyper-V host to another, without any downtime. In this section, the administrator can configure default behavior for live VM migrations to and from this specific host. This includes specifying a particular IP and associated network interface for migrations. Additional enhanced features control the type of authentication used between migration source and destination, as well as the transmission behavior.

- **Storage Migrations**

This setting allows the administrator to define the number of simultaneous storage migrations that can occur. This setting is usually best left at the default of two, unless you know your infrastructure can handle more. Most small environments will leave this at the default.

- **Replication Configuration**

Hyper-V has disaster recovery and replication built right into the product. Hyper-V replica enables replication of VMs between two or more host systems. We will be covering this in future chapters.

Chapter 3- Creating Your First Hyper-V Virtual Machine

You'll first learn the components that make up a VM, then you'll run through the process of configuring a new VM, as follows:

- Specifying name, generation, and location of the VM
- Configuring basic hardware settings for memory and networking
- Configuring and connecting storage to the VM
- Defining how the guest OS will be installed inside the VM

By the end of this chapter, you'll have a working VM whose settings you'll modify further in the next chapter. Building new VMs is a task that you'll be doing often when working with Hyper-V. Workloads change frequently, and new servers must be provisioned all the time. Learning how to properly establish a new VM will prepare you for this new, fast-changing IT landscape.

- **What Components Make a Virtual Machine?**

Before you create a new VM, it's best to understand all of the components that enable a VM to work. A virtualized OS functions in much the same manner as a physical server. It must have the same components that modern OSs depend on, just in an intangible state. This includes device drivers, BIOS, and core system devices. As there is no actual physical hardware in your VMs, virtual devices are created. The hypervisor layer provides the VM access to CPU and memory, while the parent partition provides access to networking and storage. Static items, such as the VM configuration, and file-storage related items must reside on a file system. There are several file types that exist at the VM configuration path configured in the Hyper-V host settings, as discussed in Chapter 2.

- **VHD and VHDX Files**

The storage devices attached to the VM, such as a hard drive, are virtualized also. This information is stored on physical disk via VHD or VHDX files. These files are stored at the path you defined for virtual hard disk storage in Chapter 2.

Note VHD is the older format introduced in Hyper-V 2008. It has a size limitation of 2TB. The VHDX format was introduced in Windows Server 2012, increasing the maximum size to 64TB and adding protection against storage connectivity issues and power outages. VHDX is the default format for every version of Hyper-V since 2012 R2.

- **AVHD Files**

AVHD files are very similar to VHD and VHDX files, in that they store the data contained within the VM. By default, there will be no AVHD files associated with a VM. When a checkpoint is taken, an AHVD file is created, and the VM config file is modified to redirect all incoming disk writes to it.

Creating a New Virtual Machine.

Now that you understand the underlying architecture and the files associated with a VM, you are ready to create one. Let's launch the New Virtual Machine Wizard and create a VM.

1. With Hyper-V Manager open, select the target host that will run the new VM.
2. 2. In the right-hand action pane, click New, then select Virtual Machine, as shown in Figure 3-2.

Figure 3-2. *Click New ➤ Virtual Machine to start the VM creation wizard*

3. Click the check box "Do not show this page again," to prevent the starting page from displaying again.

4. Click Next.

5. On the "specify name and location" section of the wizard, you first must define a name. This will be the VM name, as displayed inside of the Hyper-V management tools. Additionally, you can define a different location in which to store the VM. By default, this field will be populated with the paths that we defined in the Hyper-V host settings in Chapter 2.

6. Click Next

7. You'll now be on the Specify Generation screen, shown in Figure 3-3.

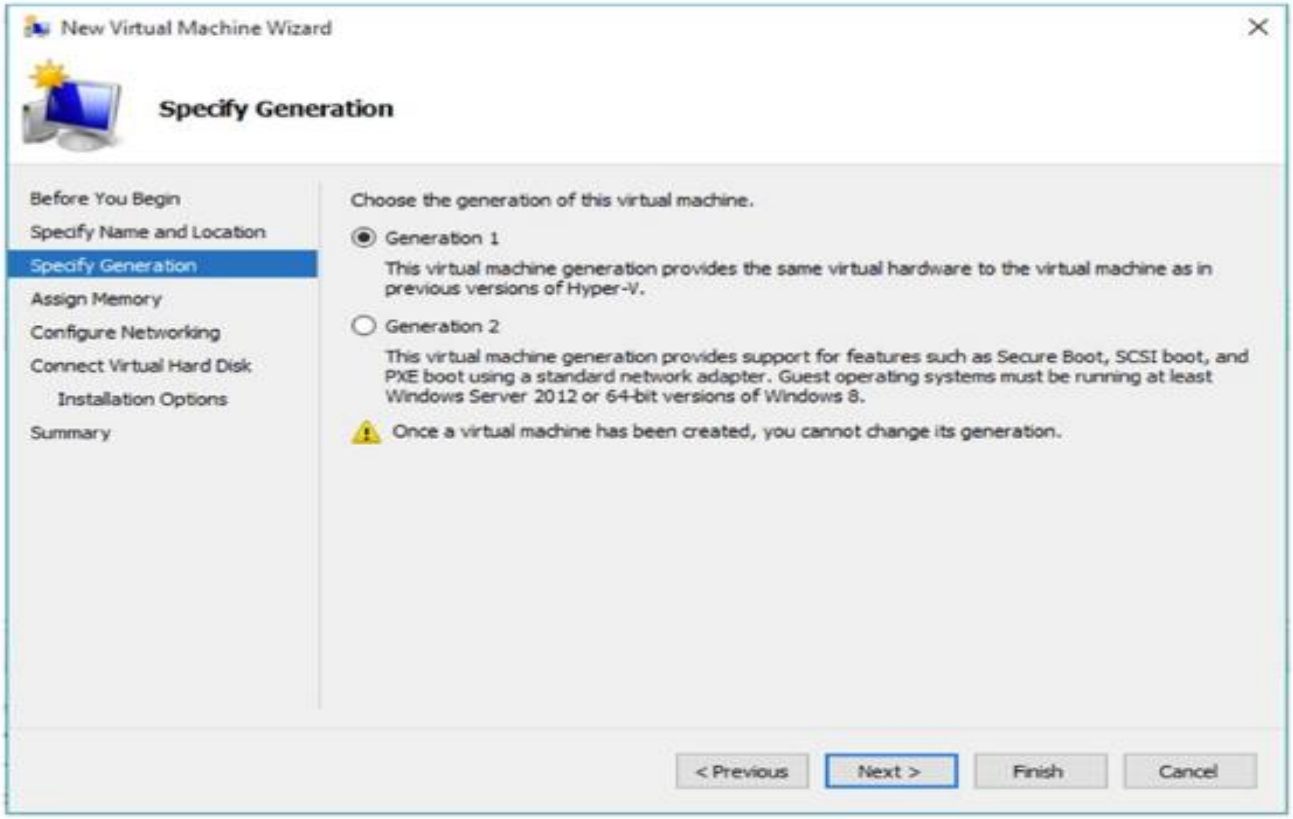


Figure 3-3. *Selecting the generation of the VM*

8. Select Generation 2.

9. Click Next.

NOTE: The VMs hosted on Hyper-V hosts prior to Server 2012 were given the designation "Generation 1" VMs. All new VMs with modern OSs running within them would be created as "Generation 2" VMs, although users still would have the option of building a VM with the older Generation 1 type, for environments that might still have some of the older Windows Server 2008/2008 R2 Hyper-V hosts in production.

A Generation 2 VM has a number of useful benefits, such as secure boot and UEFI firmware support. All new enhancements moving forward will be provided for this generation type. At the time of this writing, the following Windows OSs are supported Generation 2 OSs:

- Windows Server 2019
- Windows Server 2016
- Windows Server 2012 R2
- Windows Server 2012
- 64-bit versions of Windows 10
- 64-bit versions of Windows 8.1
- 64-bit versions of Windows 8

Caution Be very careful and sure of your decision when selecting the VM generation, because once it's configured, it remains for the life of the VM. This is one of the very few items that cannot be modified once the machine is created!

The next screen of the wizard is the memory configuration screen, as seen in Figure 3-4. This section of the wizard dictates how much memory the VM will start with and whether Dynamic Memory is enabled. Dynamic Memory is a technology in which the Hyper-V host and the guest VM work together to determine the amount of memory the VM should be assigned for optimal performance. If Dynamic Memory is left in a disabled state, the amount of memory defined in the "startup memory" field will be the static amount of memory that the VM will run with at all times.

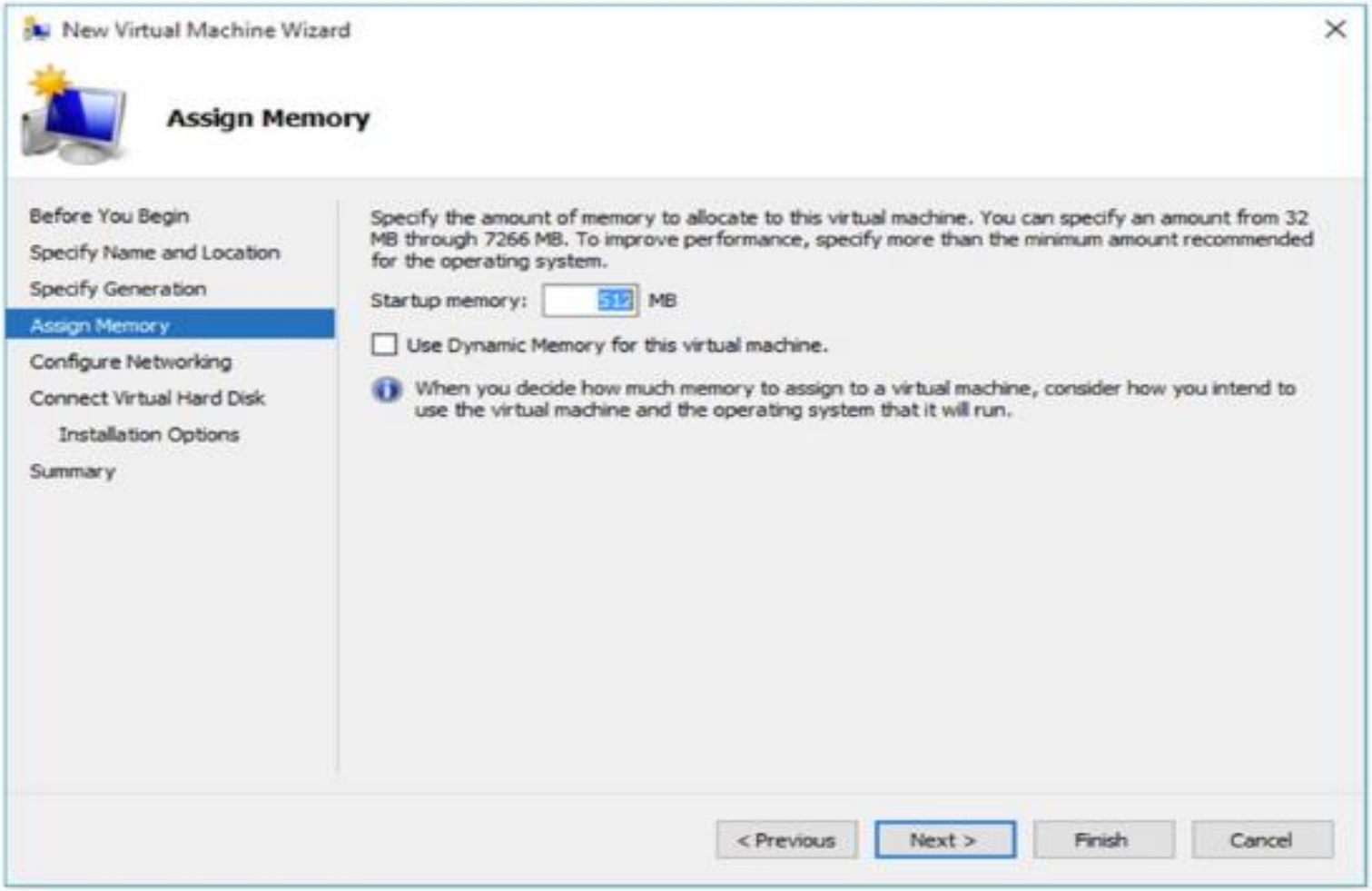


Figure 3-4. *Defining the startup memory and Dynamic Memory feature state*

- 10. Define the needed amount of memory; the default is 512MB. A minimum of 1024MB is required to install Windows Server.
- 11. Leave Dynamic Memory disabled at this time.
- 12. Click Next.

13. Leave the Connection drop-down set to "Not Connected."
14. Click Next.
15. The many storage options are shown in Figure 3-5

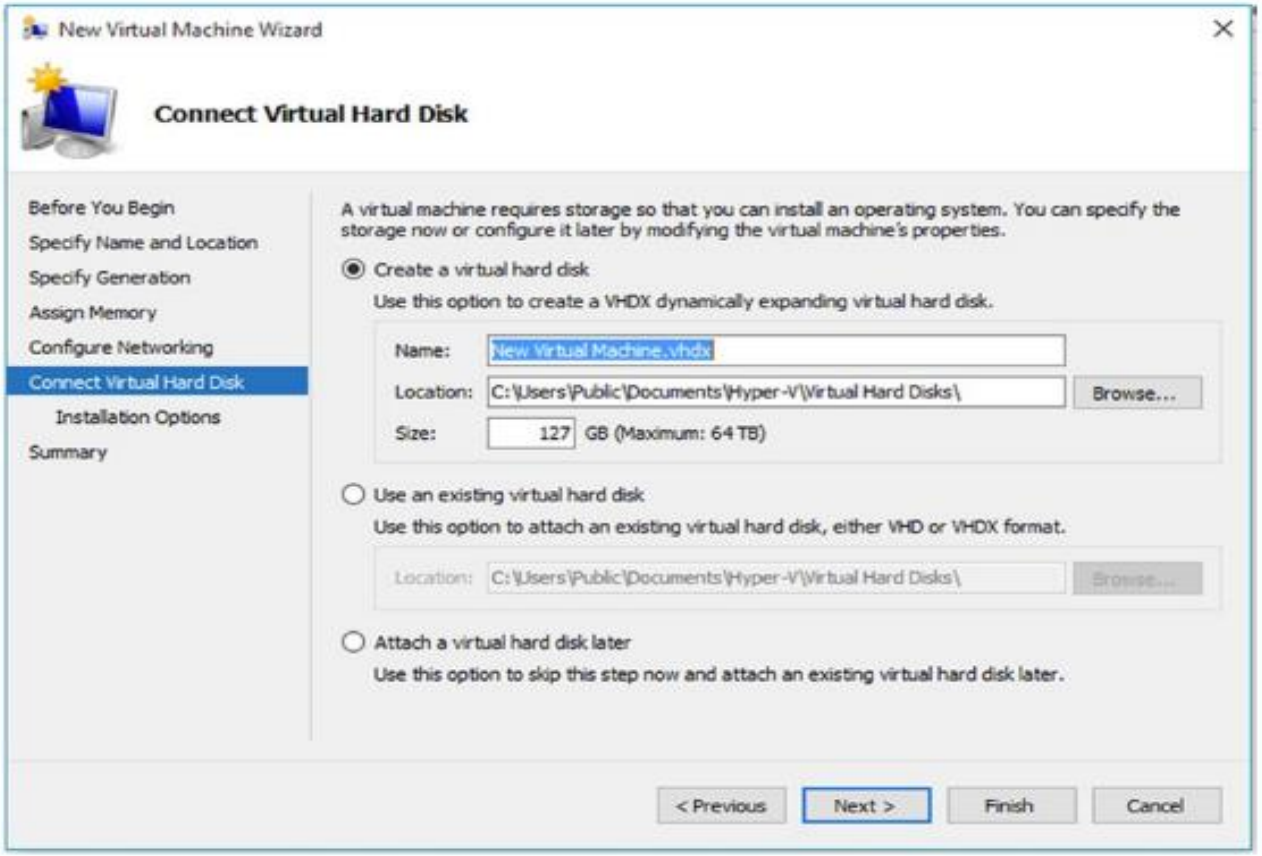


Figure 3-5. *Connect Virtual Hard Disk*

- The default option here is to create a new virtual hard disk that lets you select the size of the virtual disk and also whether you want to store it in the default virtual hard disk path. You also have the option to use a preexisting VHD or VHDX file or attach a virtual hard disk at a later time. The latter is particularly useful when you simply want to run a live OS from an ISO, such as a live Linux distribution. 16. Keep the default location for the VHDX file.
- 17. Define the size of the virtual disk you want to create (make sure you have enough space!).
- 18. Click Next
- 19. Choose installation options (see Figure 3-6).

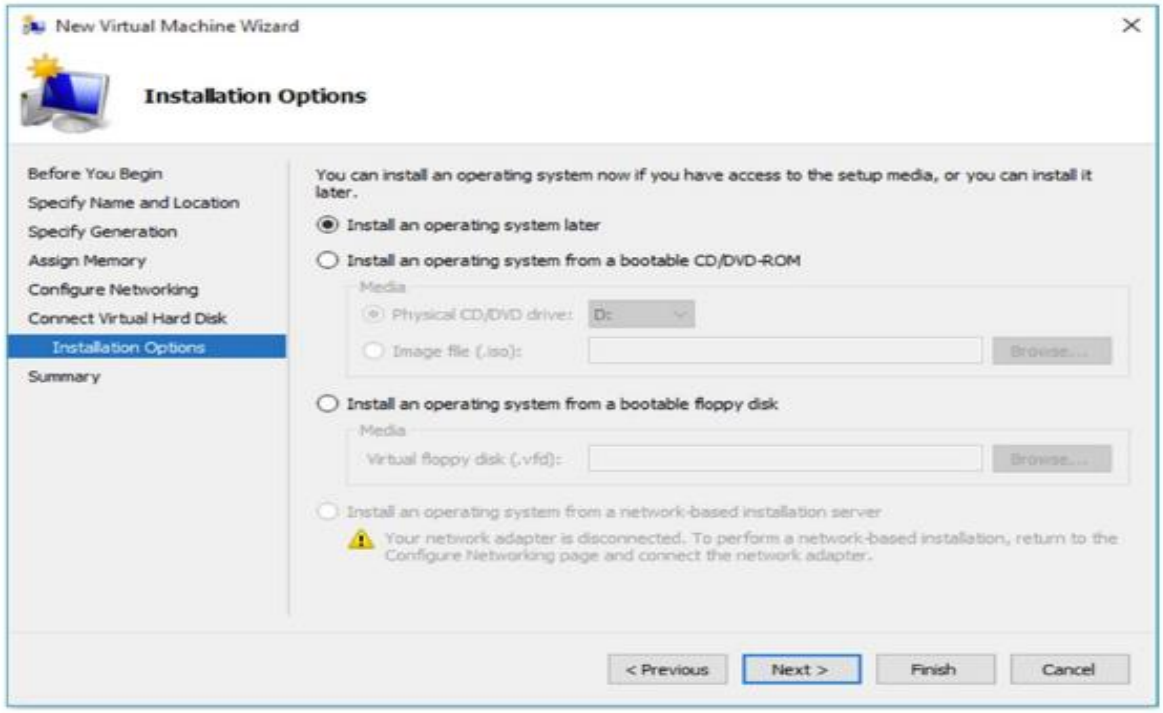


Figure 3-6. *Installation Options*

The default option is to install an OS at a later time, but if you're creating a new VM, you can't do much with it without an OS! The second option allows the VM to use the physical DVD drive of the host as an installation source. Finally, you can also use an ISO file. The hypervisor will attach the ISO to the virtual DVD drive of the new VM. You can also use a floppy disk or network source for installation of the OS, but these approaches are used less commonly.

Select Install an OS from a bootable CD/DVD-ROM.

20. Select Install an OS from a bootable CD/DVD-ROM
21. Select Image file (.iso) and navigate to your Windows Server 2016 or 2019 ISO location.
22. Click OK. Click Next
23. On the last screen of the wizard, review and confirm your settings, then click Finish.

CHAPTER 4- Introduction to the Hyper-V Extensible Switch

We're abstracting the networking layer (just like compute and storage), including the switching component. Each Hyper-V host can create one or more virtualized switching entities called a virtual switch, or vSwitch for short.

You can imagine these vSwitches as similar to a VM, in that they are intangible objects that exist within the host.

These entities are associated with the physical network cards in the host system.

The way we've always explained it to people is to imagine a normal switch and then pretend that the physical NICs on the host are uplink ports from that switch that connect back to the core networking infrastructure, as illustrated in Figure 8-1.

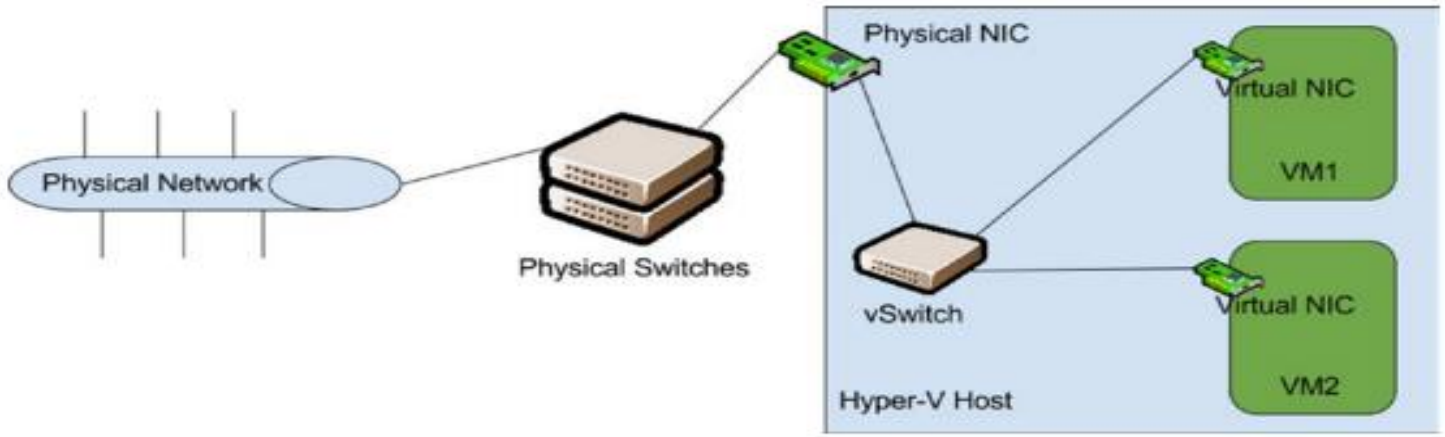


Figure 8-1. *Basic logical layout of Hyper-V networking, with connectivity to the physical network*

Virtual Switch Types in Hyper-V and Their Uses

Microsoft has provided a number of virtual switch types for use in Hyper-V environments. We have quite a bit of flexibility when it comes to controlling how vSwitches behave in Hyper-V. You have to define the switch type at the time you create it. We'll cover each type and what its primary use is.

- **External Virtual Switch**

This is the type of vSwitch you'll be creating and using 90% of the time, usually for connecting VMs to the physical network. This virtual switch type provides communication between VMs on the host and external network connectivity. Figure 8-1 is the perfect example of an external vSwitch configuration. This is the default switch type when creating a new switch, as you'll see shortly.

- **Internal Virtual Switch**

An internal virtual switch does not provide access to the physical network. This type of switch will allow VMs attached to it to talk to each other and to the host system. If you're running a lab in which you want to simulate entire environments and ensure that the test environment is isolated from the production network, you'd use an internal virtual switch to guarantee that isolation. Or, let's say you have a VM that requires additional network isolation because you're working in a high-security environment.

- **Private Virtual Switch**

This virtual switch type is exactly the same as the internal vSwitch type, except a private virtual switch will not allow the host to talk with the VMs attached to the vSwitch. You'll still be able to connect with the console to those VMs and manage them normally, but from a networking standpoint, the host will not be able to access those workloads. You would use this vSwitch type when you don't want the host itself to be able to see the IP traffic of the VMs attached to this vSwitch. You'll see this type of switch used for internal cluster traffic when we discuss clustering